

SARTHAK BISWAS

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Summary

AI Engineer focused on post-training and production ML — SFT, GRPO, DPO, RL environment design, and deep learning systems built from scratch.

Experience

AIPrep | FastAPI, PostgreSQL, Docker, S3 | Hugging Face, Unsloth AI

Remote

AI Engineer

Nov 2025 – Present

- Built an **AI video generation pipeline from 0 to 1** using **Manim** and **ElevenLabs TTS**, powering **280K+ educational videos** with multilingual narration across **14+ languages**.
- Designed **multi-stage LLM pipeline** with model routing, tool calling, and async task execution, serving **45K+ active users**.
- Built **SFT/DPO/Repair training data pipelines** using **Unsloth AI** — curated SFT, DPO, and repair datasets across 13 subjects for domain-specific model finetuning, improving model accuracy and inference latency over general-purpose LLMs.
- Deployed **RAG-powered search** and production APIs, reducing query latency and enabling contextual retrieval across educational content.
- Built a **personalized recommendation engine** using **collaborative filtering**, **pgvector** embeddings, and weighted ranking with cold-start handling.

Projects

NYC ETA Engine | *Python, PyTorch, LightGBM, MLflow, Docker*

GitHub | HF

- Built a **3-model ensemble** (neural net + LightGBM + FT-Transformer) for NYC taxi ETA prediction on **37M trips**, achieving **253s MAE** — **28% better than XGBoost baseline**, inference under 4ms on CPU.
- Built **26 features** with **Bayesian-shrunk zone-pair statistics** across **6 traffic-regime time buckets**, with a fallback hierarchy for unseen pairs — zone-pair statistics alone beat the XGBoost baseline with zero ML.
- Designed a **dual-branch embedding network** (560K params) with learned zone embeddings, implemented **FT-Transformer from scratch** (compressed 406K → 169K params), and applied **ONNX Runtime export** reducing inference from 11ms to 4ms.

Stock Trader RL Environment (OpenEnv) | *Python, FastAPI, Pydantic, Docker, HF Spaces*

GitHub

- Built for the **Meta PyTorch OpenEnv Hackathon** — qualified for the **finale (top 800 out of 32,000+ teams)** — an **OpenEnv-compliant reinforcement learning environment** for daily stock trading on 68 NIFTY stocks with real OHLCV data.
- Designed three difficulty tiers with realistic constraints (transaction costs, slippage, regime gates) and **HTTP/WebSocket interfaces** for **LLM agent** connectivity, deployed on **Hugging Face Spaces**.
- Built a **FeatureEngine** (RSI, MACD, Bollinger Bands, momentum) and a **Grader** scoring agents on Sharpe ratio, discipline, and risk management.

Autonomous Trader Agent | *Python, FastAPI, PostgreSQL, Docker, Zerodha API*

GitHub | Demo

- Built an autonomous trading system for **96 NIFTY stocks** using **cross-sectional reversal scoring** with **Information Coefficient +0.020 (large-cap), +0.025 (midcap)**.
- Engineered a **3-state regime classifier** (Bull/Neutral/Weak) with **adaptive confidence scoring**, achieving **8.6% CAGR** and **60% win rate** across 5.4 years of statistical backtesting.
- Implemented **A/B pipeline testing**, risk controls (**kill switches, regime gates, drawdown dampening**), and full state persistence to **Neon Postgres** for audit trails.

Education

Rajasthan Technical University

B.Tech in Information Technology

Kota, Rajasthan

Technical Skills

Languages: Python, TypeScript

AI / Machine Learning: PyTorch, Hugging Face Transformers, Scikit-learn, Unsloth AI, vLLM, LLMs, NLP, Reinforcement Learning

Training / Techniques: SFT, DPO, RAG, Prompt Engineering, LLM Fine-tuning, Data Pipelines

Data / Libraries: Pandas, NumPy, OpenCV, NLTK, Matplotlib

Backend / Databases: FastAPI, Celery, Redis, PostgreSQL, Vector Databases

Cloud / DevOps / MLOps: AWS/GCP, DigitalOcean, Docker, CI/CD, GitHub Actions